

Ex.No. 13: V2X Latency Comparison

Objective 1 Matlab Code and Output:

```
clc;

clear;

close all;

% Vehicle Speed (km/h)

speed = 20:20:120;

% Simulated End-to-End Latency (ms)

latency = [18 16 14 12 10 9];

disp('Vehicle Speed (km/h)   End-to-End Latency (ms)')

disp([speed' latency'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(speed,latency,'-o','LineWidth',2)

title('Vehicle Speed vs End-to-End Latency')

xlabel('Vehicle Speed (km/h)')

ylabel('Latency (ms)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

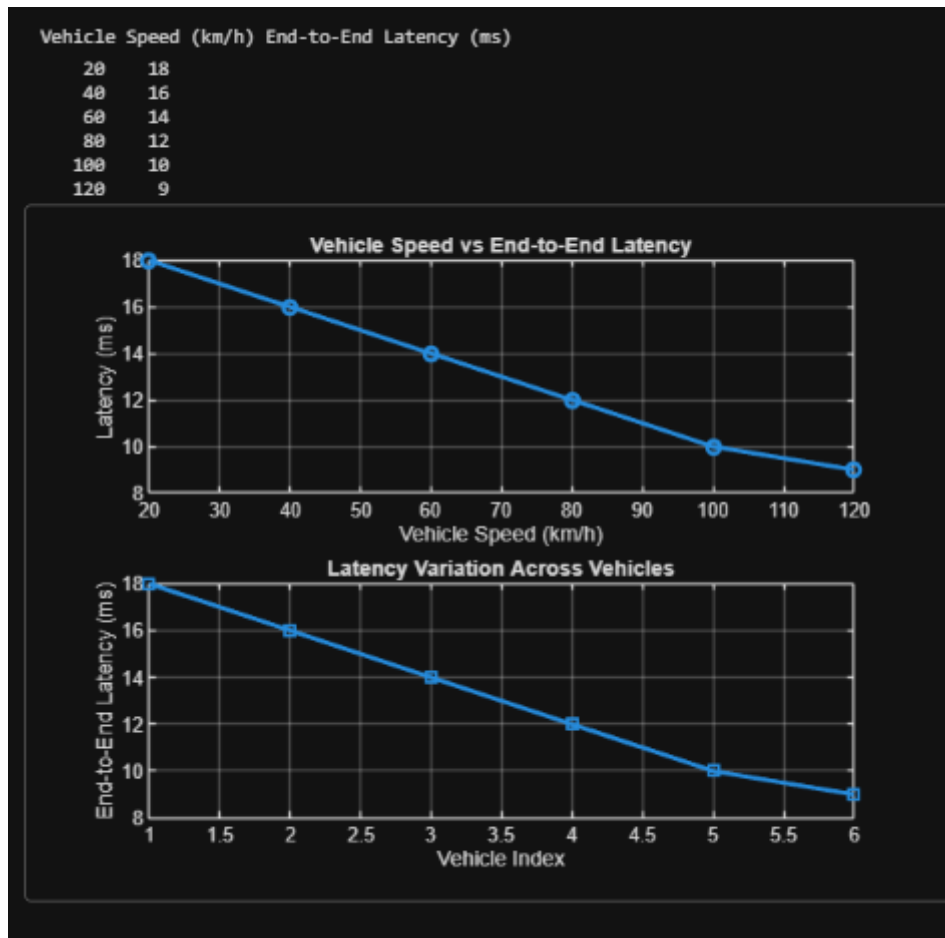
plot(1:length(latency),latency,'-s','LineWidth',2)

title('Latency Variation Across Vehicles')

xlabel('Vehicle Index')

ylabel('End-to-End Latency (ms)')

grid on
```



Objective 2 Matlab Code and Output:

```

clc;

clear;

close all;

% Vehicle Speed (km/h)
speed = 20:20:120;

% Simulated Packet Delivery Ratio (%)
PDR = [99 98 97 95 93 90];

disp('Vehicle Speed (km/h)   Packet Delivery Ratio (%)')

disp([speed' PDR'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

```

```

plot(speed,PDR,'-o','LineWidth',2)

title('Vehicle Speed vs Packet Delivery Ratio')

xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(1:length(PDR),PDR,'-s','LineWidth',2)

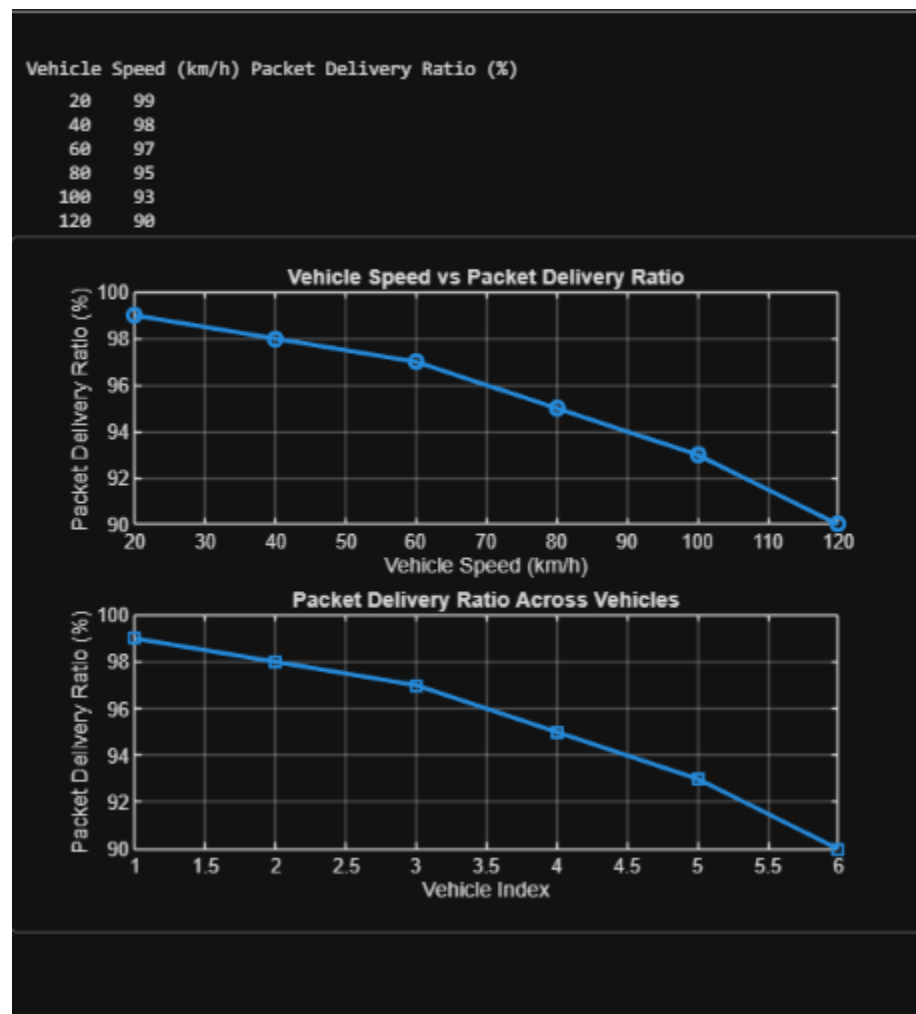
title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index')

ylabel('Packet Delivery Ratio (%)')

grid on

```



Objective 3 Matlab Code and Output:

```
clc;

clear;

close all;

% Vehicle IDs
vehicle = 1:6;

% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];

% Communication Range (m)
range = [280 260 240 220 200 180];

disp('Vehicle  Speed(km/h)  Communication Range(m)')

disp([vehicle' speed' range'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(vehicle,speed,'-o','LineWidth',2)

title('Vehicle Speed of Different Vehicles')

xlabel('Vehicle Index')

ylabel('Vehicle Speed (km/h)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(speed,range,'-s','LineWidth',2)

title('Vehicle Speed vs Communication Range')

xlabel('Vehicle Speed (km/h)')

ylabel('Communication Range (m)')

grid on
```

Vehicle Speed(km/h) Communication Range(m)

1	30	280
2	45	260
3	60	240
4	75	220
5	90	200
6	105	180

